

Non-Uniform Turán-Type problems

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Abstract

Given positive integers n, k, t , with $2 \leq k \leq n$, and $t < 2^k$, let $m(n, k, t)$ be the minimum size of a family $\mathcal{F}$ of nonempty subsets of $[n]$ such that every k -subset of $[n]$ contains at least t members of $\mathcal{F}$, and every $(k-1)$ -subset of $[n]$ contains at most $t-1$ members of $\mathcal{F}$. For fixed k and t , we determine the order of magnitude of $m(n, k, t)$. This extends a result of Sloan et. al [7]. For fixed k , our results imply that $M(k, t) = \lim_{n \rightarrow \infty} \log_n m(n, k, t)$ is not monotone in t . This is related to a question of Füredi et. al [5].

We also consider the Turán number $T_{\geq r}(n, k, t)$ defined as the minimum size of a family $\mathcal{F} \subset \binom{[n]}{\geq r}$ such that every k -subset of $[n]$ contains at least t members of $\mathcal{F}$. Our results include proving that $T_{\geq r}(n, k, t) = (1 + o(1))T_r(n, k, t)$ for fixed r, k, t with $t \leq \binom{k}{r}$ and $n \rightarrow \infty$.

1 Introduction

In this paper we study two extremal functions for set systems, the threshold function $m(n, k, t)$ and the non-uniform Turán number $T_{\geq r}(n, k, t)$.

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1.1 $m(n, k, t)$

Given integers n, k, t , with $2 \leq k \leq n$, and $t < 2^k$, a family $\mathcal{F} \subset 2^{[n]} \setminus \emptyset$ has the (k, t) -threshold property if every k -subset of $[n]$ contains at least t sets from $\mathcal{F}$, and every $(k-1)$ -subset of $[n]$ contains at most $t-1$ sets from $\mathcal{F}$. Let $m(n, k, t)$ denote the minimum size of a family of subsets of $[n]$ with the (k, t) -threshold property. This function first arose in problems on computer science (although the notation $m(n, k, t)$ was not used until [5]). Jukna [6] studied it in circuit complexity theory when considering the trade-offs in threshold circuits. Sloan-Takata-Turán [7] studied $m(n, k, t)$ in the context of *frequent sets* of Boolean matrices, a concept used in the algorithmic, combinatorial approach to knowledge discovery and data mining [1]. It was shown in [7] that $m(n, k, t) = \Theta(n^{k-1})$ for $1 < t < k$ and $m(n, 3, 2) = \binom{n-1}{2} + 1$.

Füredi-Sloan-Takata-Turán [5] studied $m(n, k, t)$ as a purely combinatorial problem. They proved that for fixed k , $m(n, k, 2) = (1 + o(1))T_{k-1}(n, k, 2)$, where $T_{k-1}(n, k, 2)$ is a generalized Turán number. In [5], the smallest nontrivial case, $m(n, 4, 3)$, is determined exactly for large n . It is also mentioned in [5] that one expects if n and k are fixed, then $m(n, k, t)$ decreases with t for $t < k$.

We study $m(n, k, t)$ for all $1 \leq t < 2^k$. Our first result not only shows that $m(n, k, t)$ exists for every t , but also determines its order of magnitude. In particular, it implies that for fixed k and t , $\log_n m(n, k, t)$ is far from being monotone in t for $1 \leq t < 2^k$.

For natural numbers $b \leq a$, we write $(b, a]$ for the interval $\{b+1, \dots, a\}$. Let $\binom{a}{\leq b} = \sum_{i=1}^b \binom{a}{i}$ and $\binom{a}{0} = 1$. For $1 \leq j \leq k$, let $I_j = ((\binom{k}{\leq j-1}), \binom{k}{\leq j}]$. Note that $I_1, I_2, \dots, I_k$ partition $(1, 2^k - 1]$.

Theorem 1. *Let $2 \leq k \leq n$ and $2 \leq t \leq 2^k - 1$. Define j by $t \in I_j$.*

- *If $\binom{k-1}{j} \notin I_j$, then $\frac{1}{\binom{k}{j}} \binom{n}{j} \leq m(n, k, t) \leq \binom{n}{\leq j}$.*
- *If $\binom{k-1}{j} \in I_j$, then*

$$\begin{aligned} c \binom{n}{k-j} &\leq m(n, k, t) \leq \binom{n}{k-j} \quad \text{for} \quad \binom{k}{\leq j-1} < t \leq \binom{k-1}{j}, \\ \frac{1}{\binom{k}{j}} \binom{n}{j} &\leq m(n, k, t) \leq \binom{n}{\leq j} \quad \text{for} \quad \binom{k-1}{j} < t \leq \binom{k}{\leq j}. \end{aligned} \tag{1}$$

where $c > 0$ is a constant that depends only on k and j .

Remark:

1. When $j = 1$, we have $k-1 \in (1, k] = I_1$. Therefore (1) reduces to $m(n, k, t) = \Theta(n^{k-1})$ for $2 \leq t \leq k-1$, a result that was proved in [7].
2. The lower bound in (1) is trivial unless n is large since the constant c is very small. Actually, we obtain c from iterated Ramsey numbers: $c = 1/(2m_1)^{k-j}$, where $m_\ell = R^{k-j-\ell}(k-1, m_{\ell+1})$ for $\ell = 1, \dots, k-2j$ and $m_{k-2j+1} = k$.
3. The case $t = 1$ is covered later by Proposition 3.

Define

$$M(k, t) = \lim_{n \rightarrow \infty} \log_n m(n, k, t).$$

Theorem 1 implies that $M(k, t)$ exists for all $2 \leq t < 2^t$ and gives the values of $M(k, t)$ as follows.

Corollary 2. *Let I_j be defined as in Theorem 1. If $\binom{k-1}{j} \notin I_j$, then $M(k, t) = j$ for $t \in I_j$. Otherwise,*

$$M(k, t) = \begin{cases} k-j & \text{if } \binom{k}{\leq j-1} < t \leq \binom{k-1}{j}, \\ j & \text{if } \binom{k-1}{j} < t \leq \binom{k}{\leq j}. \end{cases}$$

Corollary 2 implies that $M(k, t)$ is far from being monotone in t . For example, the following table of $m(n, 5, t)$ illustrates this phenomenon.

$t =$	1	2 - 4	5	6	7 - 15	16 - 25	26 - 30	31
$m(n, 5, t) \approx$	n^5	n^4	n	n^3	n^2	n^3	n^4	n^5

For the following special values of t , it is fairly straightforward to determine $m(n, k, t)$ exactly.

Proposition 3.

$$m(n, k, t) = \begin{cases} \binom{n}{k} & t = 1 \\ n & t = k \\ \binom{n}{\leq k-1} & t = 2^k - 2 \\ \binom{n}{\leq k} & t = 2^k - 1 \end{cases}$$

1.2 $T_{\geq r}(n, k, t)$

In the definition of $m(n, k, t)$, if we remove the second restriction for the set system, we obtain another extremal function which appears not to have been studied before, *the nonuniform Turán number* $T_{\geq 1}(n, k, t)$. More generally, given integers n, k, t , with $1 \leq r \leq k \leq n$, and $0 \leq t < 2^k$, a family $\mathcal{F} \in \binom{[n]}{\geq r}$ is called a Turán- $\geq r(n, k, t)$ -system if every k -subset of $[n]$ contains at least t members of $\mathcal{F}$. The (non-uniform) Turán number $T_{\geq r}(n, k, t)$ is defined as the minimum size of a Turán- $\geq r(n, k, t)$ -system. When $t \leq \binom{k}{r}$, replacing the subscript $\geq r$ by r , we obtain the well-known (uniform) Turán number $T_r(n, k, t) = \binom{n}{r} - \text{ex}(n, \mathcal{G})$, where $\mathcal{G}$ is the family of the k -vertex r -graphs with at least $\binom{k}{r} - t + 1$ edges. For example, the graph Turán number $T_2(n, k, t)$ can be found (in many cases) by Turán's Theorem [8] and the Erdős-Stone-Simonovits Theorem [2, 4]. However, the asymptotics of $T_r(n, k, t)$ are not known for any $t \geq 1$ when $r \geq 3$. For example, determining $T_3(n, 4, 1)$ is a well-known conjecture of Turán.

The second part of this paper is on the relationship between $T_{\geq r}(n, k, t)$ and $T_r(n, k, t)$. Often nonuniform statements of classical extremal results are much easier to prove than their uniform analogues (the Erdős-Ko-Rado theorem is one example); our results show that this is not the case for the nonuniform Turán number.

Observation 1. Let $1 \leq r \leq k$ and $0 \leq t \leq \sum_{i=r}^k \binom{k}{i}$. Define j by $\sum_{i=r}^{j-1} \binom{k}{i} \leq t < \sum_{i=r}^j \binom{k}{i}$ and let $t_0 = t - \sum_{i=r}^{j-1} \binom{k}{i}$.

If $\mathcal{F}$ is a Turán- $j(n, k, t_0)$ -system, then $\mathcal{F}' = \bigcup_{i=r}^{j-1} \binom{[n]}{i} \cup \mathcal{F}$ is a Turán- $\geq r(n, k, t)$ -system. This implies that

$$T_{\geq r}(n, k, t) \leq \sum_{i=r}^{j-1} \binom{n}{i} + T_j(n, k, t_0). \quad (2)$$

The following theorem shows that equality holds asymptotically in (2) (the special case $j = r$ is stated in the abstract).

Theorem 4. Let r, k, t, j, t_0 be as in Observation 1.

1. If $t_0 = 0$, then $T_{\geq r}(n, k, t) = \sum_{i=r}^{j-1} \binom{n}{i}$.

2. If $t_0 \neq 0$, then as $n \rightarrow \infty$,

$$T_{\geq r}(n, k, t) = (1 + o(1)) \left(\sum_{i=r}^{j-1} \binom{n}{i} + T_j(n, k, t_0) \right).$$

We conjecture that (2) is sharp.

Conjecture 5. *Let r, k, t, j, t_0 be as in Observation 1. Then*

$$T_{\geq r}(n, k, t) = \sum_{i=r}^{j-1} \binom{n}{i} + T_j(n, k, t_0).$$

We also have the following partial results.

Proposition 6. *Conjecture 5 holds in the following cases:*

- $t \leq r + 1$
- $r = 1, t \leq k$

Furthermore, $T_{\geq 1}(n, k, t) = n + T_{\geq 2}(n, k, t - k)$ for every $t > k$.

This paper is organized as follows. In Section 2 we present the proof of Theorem 1 and Propositions 3. In Section 3, we prove the results on $T_{\geq r}(n, k, t)$.

Most of our notations are standard: $[n] = \{1, 2, \dots, n\}$. For a set system $\mathcal{F}$, let $\mathcal{F}_t$ denote the family of t -sets in $\mathcal{F}$, let $\mathcal{F}_{\leq t} = \cup_{i \leq t} \mathcal{F}_i$ and $\mathcal{F}_{\geq t} = \cup_{i \geq t} \mathcal{F}_i$. Given a set X and an integer a , let $2^X = \{S : S \subseteq X\}$, $\binom{X}{a} = \{S \subset X : |S| = a\}$, $\binom{X}{\leq a} = \{S \subset X : 1 \leq |S| \leq a\}$ and $\binom{X}{\geq a} = \{S \subset X : |S| \geq a\}$. We write $\mathcal{F}(X)$ for $\mathcal{F} \cap 2^X$. An r -graph on X is a (hyper)graph whose edges are r -subsets of X .

2 Proofs of results on $m(n, k, t)$

A family $\mathcal{F} \subseteq 2^{[n]} \setminus \emptyset$ is called a (k, t) -system if $\mathcal{F}$ has the (k, t) -threshold property, *i.e.*, it satisfies the following two conditions:

Property D (DENSE): Every k -set in $[n]$ contains at least t sets from $\mathcal{F}$,

Property S (SPARSE): Every $(k - 1)$ -set in $[n]$ contains at most $t - 1$ sets from $\mathcal{F}$.

Our proof of Theorem 1 is facilitated by a sequence of four lemmas, which give upper and lower bounds for $m(n, k, t)$ in various ranges of t .

Lemma 7. *If $\binom{k-1}{j-1} < t \leq \binom{k}{j}$, then $m(n, k, t) \leq \binom{n}{k-j}$.*

Proof. Let $\mathcal{F} = \binom{[n]}{k-j}$. Then every k -set in $[n]$ contains $\binom{k}{k-j} = \binom{k}{j} \geq t$ members of $\mathcal{F}$; every $(k-1)$ -set in $[n]$ contains $\binom{k-1}{k-j} = \binom{k-1}{j-1} < t$ members of $\mathcal{F}$. Hence, $\mathcal{F}$ satisfies Property **D** and **S**, in other words, $\mathcal{F}$ is a (k, t) -system. Consequently, $m(n, k, t) \leq |\mathcal{F}| = \binom{n}{k-j}$. $\square$

Lemma 8. *If $\binom{k-1}{j} < t \leq \binom{k}{\leq j}$, then $m(n, k, t) \leq \binom{n}{\leq j}$.*

Proof. Consider the smallest $i' \in [1, j]$ and the largest $i \in [1, j]$ such that

$$1 + \sum_{\ell=i'}^j \binom{k-1}{\ell} \leq t \leq \sum_{\ell=i}^j \binom{k}{\ell}.$$

We first show that $i' \leq i$. When $i = j$, this is obviously true. So we may assume that $i < j$. The choice of i implies that

$$\begin{aligned} t &> \sum_{\ell=i+1}^j \binom{k}{\ell} = \binom{k}{i+1} + \sum_{\ell=i+2}^j \binom{k}{\ell} \\ &\geq \left[\binom{k-1}{i} + \binom{k-1}{i+1} \right] + \sum_{\ell=i+2}^j \left[\binom{k-1}{\ell} + 1 \right] \\ &\geq \sum_{\ell=i}^j \binom{k-1}{\ell} + 1, \end{aligned} \tag{3}$$

where the second inequality follows from the fact $\binom{k}{i+1} = \binom{k-1}{i} + \binom{k-1}{i+1}$ and $\binom{k}{\ell} \geq \binom{k-1}{\ell} + 1$ for every $\ell < k$. The choice of i' and (3) thus imply that $i' \leq i$.

Now let $\mathcal{F} = \cup_{\ell=i}^j \binom{[n]}{\ell}$. Every k -set of $[n]$ has $\sum_{\ell=i}^j \binom{k}{\ell} \geq t$ members of $\mathcal{F}$; every $(k-1)$ -set of $[n]$ has $\sum_{\ell=i}^j \binom{k-1}{\ell} \leq \sum_{\ell=i'}^j \binom{k-1}{\ell} \leq t-1$ members of $\mathcal{F}$. Consequently, $\mathcal{F}$ is a (k, t) -system and $m(n, k, t) \leq |\mathcal{F}| = \sum_{\ell=i}^j \binom{n}{\ell} \leq \binom{n}{\leq j}$. $\square$

Lemma 9. *If $t > \binom{k}{\leq j-1}$, then $m(n, k, t) \geq \frac{\binom{n}{j}}{\binom{k}{j}}$.*

Proof. Suppose that $t > \binom{k}{\leq j-1}$ and $\mathcal{F}$ is a (k, t) -system. By Property **D**, every k -set in $[n]$ must contain at least one set $S \in \mathcal{F}_{\geq j}$. Since each such S lies in at most $\binom{n-|S|}{k-|S|} \leq \binom{n-j}{k-j}$ k -sets, we have $|\mathcal{F}| \geq \binom{n}{k} / \binom{n-j}{k-j} = \binom{n}{j} / \binom{k}{j}$. $\square$

Lemma 10. *If $\binom{k}{\leq j-1} < t \leq \binom{k-1}{j}$, then $m(n, k, t) \geq c \binom{n}{k-j}$, where c is a constant which depends only on k and j .*

Proof. First, the assumption $\binom{k}{\leq j-1} < \binom{k-1}{j}$ implies that $j < k - j$. Let $\mathcal{F}$ be a (k, t) -system. We claim that $\mathcal{F}$ contain no copy of $K_{k-1}^{(i)}$, the complete i -graph of order $k - 1$ for every $i \in [j, k - 1 - j]$. Otherwise, we obtain a $(k - 1)$ -set which contains at least $\binom{k-1}{i} \geq \binom{k-1}{j} \geq t$ members of $\mathcal{F}$, a contradiction.

Recall that the Ramsey number $R^{(i)}(s, t)$ is the smallest N such that every i -graph on N vertices contains a copy of either $K^{(i)}(s)$ or $\bar{K}^{(i)}(t)$. Define $m_\ell = R^{k-j-\ell}(k - 1, m_{\ell+1})$ for $\ell = 1, \dots, k - 2j$ and $m_{k-2j+1} = k$. Consider an m_1 -set S_1 . Because $K_{k-1}^{(k-j-1)} \not\subseteq \mathcal{F}$, the definition of m_1 implies that there exists a m_2 -subset $S_2 \subseteq S_1$ with all of its $(k - j - 1)$ -subsets absent from $\mathcal{F}$. Repeating this analysis, we find a sequence of subsets $S_3 \supseteq \dots \supseteq S_{k-2j+1} = S$ of sizes $m_3 > \dots > m_{k-2j+1} = k$, respectively. The k -set S thus contains no members of $\mathcal{F}$ of size $k - j - 1, \dots, j$. On the other hand, by Property **D**, the k -set S must contain at least $t > \binom{k}{\leq j-1}$ members of $\mathcal{F}$, thus at least one member of $\mathcal{F}_{\geq j}$. Since S contains no ℓ -sets with $j \leq \ell \leq k - j - 1$ from $\mathcal{F}$, we conclude that S contains a member of $\mathcal{F}_{\geq k-j}$. This means that every m_1 -set of $[n]$ contains at least one member of $\mathcal{F}_{\geq k-j}$. Consequently,

$$|\mathcal{F}| \geq \frac{\binom{n}{m_1}}{\binom{n-k+j}{m_1-k+j}} = \frac{\binom{n}{k-j}}{\binom{m_1}{k-j}} = c \binom{n}{k-j}. \quad \square$$

Proof of Theorem 1. Recall that $I_j = ((\binom{k}{\leq j-1}), \binom{k}{j})$ for $j \in [k]$.

If $\binom{k-1}{j} \notin I_j$, then by Lemma 8 and Lemma 9, for every $t \in I_j$, we have

$$\frac{\binom{n}{j}}{\binom{k}{j}} \leq m(n, k, t) \leq \binom{n}{\leq j}.$$

If $\binom{k-1}{j} \in I_j$, then we partition I_j into I_j^1 and I_j^2 , where $I_j^1 = ((\binom{k}{\leq j-1}), \binom{k-1}{j})$ and $I_j^2 = ((\binom{k-1}{j}), \binom{k}{j})$. When $t \in I_j^2$, again by Lemma 8 and Lemma 9, we have

$$\frac{\binom{n}{j}}{\binom{k}{j}} \leq m(n, k, t) \leq \binom{n}{\leq j}$$

Otherwise, $t \in I_j^1$, Lemma 7 and Lemma 10 imply

$$c \binom{n}{k-j} \leq m(n, k, t) \leq \binom{n}{k-j}. \quad \square$$

Before proving Propositions 3, we make the following observation on the structure of a minimal (k, t) -system for large t .

Observation 2. Let $\mathcal{F}$ be a minimal (k, t) -system for some $t \geq 2^{k-1}$. Then we may assume that

$$A \in \mathcal{F} \Rightarrow 2^A \subset \mathcal{F}. \quad (4)$$

In fact, if $A \in \mathcal{F}$ contains a nonempty subset $B \notin \mathcal{F}$, then we replace A by B in $\mathcal{F}$, the new set system $\mathcal{F}'$ clearly satisfies Property **D** as well. Since $t > 2^{k-1} - 1$, Property **S** also holds in $\mathcal{F}'$ (any $(k-1)$ -set of $[n]$ has at most $2^{k-1} - 1 < t$ nonempty subsets). We repeat this process until (4) holds in the resulting set system.

We are ready to prove Proposition 3.

Proof of Proposition 3. Trivially $m(n, k, 1) = \binom{n}{k}$ and $m(n, k, 2^k - 1) = \binom{n}{\leq k}$.

- **t = k:** $m(n, k, k) \leq n$, because $\binom{[n]}{1}$ is a (k, k) -system. On the other hand, if $\mathcal{F}$ is a (k, k) -system, then every k -set in $[n]$ contains k members of $\mathcal{F}$. Since each member of $\mathcal{F}$ lies in at most $\binom{n-1}{k-1}$ k -sets (with equality for a singleton), $\mathcal{F} \geq k \binom{n}{k} / \binom{n-1}{k-1} = n$. Hence, $m(n, k, k) = n$.
- **t = $2^k - 2$:** Consider a minimal $(k, 2^k - 2)$ -system $\mathcal{F}$. The assumption $k \geq 2$ implies that $t = 2^k - 2 > 2^{k-1} - 1$, thus (4) applies to $\mathcal{F}$. We claim that $\mathcal{F}_k = \emptyset$. Suppose instead, that there exists a set $S \in \mathcal{F}_k$. Then S contains $2^k - 1$ members of $\mathcal{F}$ by (4). After we remove S from $\mathcal{F}$, the new system is still a $(k, 2^k - 2)$ -system, which contradicts the minimality of $\mathcal{F}$. Hence $\mathcal{F}_k = \emptyset$, or $\mathcal{F} \subseteq \binom{[n]}{\leq k-1}$. Since $t = 2^k - 2$, Property **D** forces $\mathcal{F} = \binom{[n]}{\leq k-1}$. Therefore, $m(n, k, 2^k - 2) = |\mathcal{F}| = \binom{n}{\leq k-1}$. $\square$

3 Proof of results on $T_{\geq r}(n, k, t)$

Recall that $\mathcal{F}(S) = \mathcal{F} \cap 2^S$ for a set system $\mathcal{F}$ and a set S .

To prove Theorem 4, because of Observation 1, it suffices to verify the corresponding lower bounds for $T_{\geq r}(n, k, t)$.

Proof of Theorem 4 Part I. Let $\mathcal{F} \subset \binom{[n]}{\geq r}$ be a minimal Turán- $_{\geq r}(n, k, t)$ -system. We are to show that $|\mathcal{F}| \geq \sum_{i=r}^{j-1} \binom{n}{i}$. Consider $\overline{\mathcal{F}_{< j}} = \bigcup_{i=r}^{j-1} \binom{[n]}{i} \setminus \mathcal{F}$. For every k -set S of $[n]$,

$$\sum_{i=r}^{j-1} \binom{k}{i} = t \leq |\mathcal{F}(S)| = |\mathcal{F}_{< j}(S)| + |\mathcal{F}_{\geq j}(S)| = \sum_{i=r}^{j-1} \binom{k}{i} - |\overline{\mathcal{F}_{< j}}(S)| + |\mathcal{F}_{\geq j}(S)|.$$

Therefore $|\overline{\mathcal{F}_{< j}}(S)| \leq |\mathcal{F}_{\geq j}(S)|$. Consequently (also using the fact that $\binom{n-x}{k-x}$ is a decreasing function of x for $0 \leq x \leq k$),

$$|\overline{\mathcal{F}_{< j}}| \binom{n-j}{k-j} < \sum_{S \in \binom{[n]}{k}} |\overline{\mathcal{F}_{< j}}(S)| \leq \sum_{S \in \binom{[n]}{k}} |\mathcal{F}_{\geq j}(S)| \leq |\mathcal{F}_{\geq j}| \binom{n-j}{k-j}.$$

This implies that $|\overline{\mathcal{F}_{< j}}| \leq |\mathcal{F}_{\geq j}|$. Therefore

$$|\mathcal{F}| = |\mathcal{F}_{< j}| + |\mathcal{F}_{\geq j}| \geq |\mathcal{F}_{< j}| + |\overline{\mathcal{F}_{< j}}| = \sum_{i=r}^{j-1} \binom{n}{i}. \quad \square$$

The main tool to prove the second part of Theorem 4 is the following theorem of Erdős and Simonovits, which proves a phenomenon called supersaturation.

Theorem 11 (Erdős-Simonovits [3]). *For every $\varepsilon > 0$ and a family of r -graphs $\mathcal{G}$ on k vertices, there exists a $\delta > 0$, such that every r -graph on n vertices with at least $(1 + \varepsilon)\text{ex}(n, \mathcal{G})$ edges contains at least $\delta \binom{n}{k}$ copies of members of $\mathcal{G}$.*

Proof of Theorem 4 Part II. We are to show that for any $\varepsilon > 0$, there exists an integer $n_0 > 0$, such that for any $n \geq n_0$,

$$T_{\geq r}(n, k, t) \geq (1 - \varepsilon) \left(\sum_{i=r}^{j-1} \binom{n}{i} + T_j(n, k, t_0) \right). \quad (5)$$

Since $\sum_{i=r}^{j-1} \binom{n}{i} = O(n^{j-1})$, (5) is a consequence of the following claims:

1. $T_{\geq r}(n, k, t) \geq T_{\geq j}(n, k, t_0)$,
2. $T_{\geq j}(n, k, t_0) = \Omega(n^j)$,
3. $T_{\geq j}(n, k, t_0) > (1 - \varepsilon/2)T_j(n, k, t_0)$.

To see Claim 1, let $\mathcal{F} \subset \binom{[n]}{\geq r}$ be a minimal Turán- $\geq_r(n, k, t)$ -system. By definition, every k -set of $[n]$ contains at least t members of $\mathcal{F}$. Since $t_0 = t - \sum_{i=r}^{j-1} \binom{k}{i}$, every k -set of $[n]$ contains at least t_0 sets from $\mathcal{F}$ of cardinality at least j .

Claim 2 follows from an averaging argument (and $t_0 > 0$),

$$|T_{\geq j}(n, k, t_0)| \geq t_0 \binom{n}{k} / \binom{n-j}{k-j} = t_0 \binom{n}{j} / \binom{k}{j} = \Omega(n^j).$$

Now we prove Claim 3. Suppose that $\mathcal{F}$ is a minimal $\text{Turán}_{\geq j}(n, k, t_0)$ -system. It suffices to show that $\mathcal{F}_j > (1 - \varepsilon)T_j(n, k, t_0)$. Suppose instead, that $\mathcal{F}_j \leq (1 - \varepsilon/2)T_j(n, k, t_0)$, or $|\overline{\mathcal{F}_j}| \geq (1 + \varepsilon/2)\text{ex}(n, \mathcal{G})$, where $\mathcal{G}$ is the family of all j -graphs on k vertices with more than $\binom{k}{2} - t_0$ edges. By Theorem 11 (applied with input $\varepsilon/2$), the j -graph with vertex set $[n]$ and edge set $\overline{\mathcal{F}_j}$ contains at least $\delta\binom{n}{k}$ copies of (not necessarily the same) members of $\mathcal{G}$. In other words, there are at least $\delta\binom{n}{k}$ k -sets of $[n]$ that contain fewer than t_0 members of $\mathcal{F}_j$.

Now consider the family of k -sets of $[n]$ which contains at least one member of $\mathcal{F}_i$ for some $i > j$. Denote this by $\mathcal{K}_i$ and let $\mathcal{K} = \cup_{j < i \leq k} \mathcal{K}_i$. Since $|\mathcal{K}_i| \leq |\mathcal{F}_i|\binom{n-i}{k-i}$ and $|\mathcal{F}| \leq \binom{n}{j}$,

$$|\mathcal{K}| = \sum_{j < i \leq k} |\mathcal{K}_i| \leq \sum_{j < i \leq k} |\mathcal{F}_i|\binom{n-i}{k-i} \leq \binom{n-j-1}{k-j-1} |\mathcal{F}| \leq \binom{n-j-1}{k-j-1} \binom{n}{j} < n^{k-1}.$$

Since $\delta\binom{n}{k} > n^{k-1} > |\mathcal{K}|$, at least one k -set S of $[n]$ contains fewer than t_0 members of $\mathcal{F}_j$ and no member of $\mathcal{F}_i$ for $i > j$. Consequently S contains fewer than t_0 members of $\mathcal{F}$. This contradicts the assumption that $\mathcal{F}$ is a $\text{Turán}_{\geq j}(n, k, t_0)$ -system. $\square$

The following observation is needed for the proof of Proposition 6.

Observation 3. *Suppose that $\mathcal{F}$ is a minimal $\text{Turán}_{\geq r}(n, k, t)$ -system. We may assume that*

$$A \in \mathcal{F} \Rightarrow \binom{A}{\geq r} \subset \mathcal{F}. \quad (6)$$

In fact, if there exist a set $A \in \mathcal{F}$ which has a subset $B \in \binom{[n]}{\geq r} \setminus \mathcal{F}$, we may replace A by B in $\mathcal{F}$ and the new set system is still a $\text{Turán}_{\geq r}(n, k, t)$ -system.

Proof of Proposition 6. First, let $\mathcal{F}$ be a minimal $\text{Turán}_{\geq r}(n, k, t)$ -system satisfying (6). When $t \leq r + 1$, we claim that there is no set $A \in \mathcal{F}_{>r}$ (Conjecture 5 thus holds). Otherwise, (6) implies that $|\mathcal{F}(A)| \geq 1 + r + 1 \geq t + 1$ for such A . We may therefore remove A from $\mathcal{F}$ and the new set system is still a $\text{Turán}_{\geq r}(n, k, t)$ -system (thus $\mathcal{F}$ is not minimal).

Second, we are to show $T_{\geq 1}(n, k, t) = T_1(n, k, t) = n - k + t$ when $t \leq k$. Clearly a set of $n - k + t$ singletons is a $\text{Turán}_{\geq 1}(n, k, t)$ -system. To show $T_{\geq 1}(n, k, t) \geq n - k + t$ for $t \leq k$, we use induction on k . When $k = 1$, $t = 1$ and $T_{\geq 1}(n, 1, 1) = n$ trivially. Suppose the statement holds for $k - 1$. Let $\mathcal{F}$ be a minimal $\text{Turán}_{\geq 1}(n, k, t)$ -system satisfying (6).

Suppose that there is $a \in [n]$ such that $\{a\} \notin \mathcal{F}$. By (6), we know that no member of $\mathcal{F}$ contains a , and $\mathcal{F}$ is therefore a $\text{Turán-}_{\geq 1}(n-1, k-1, t)$ -system on $[n] \setminus \{a\}$. Consequently,

$$|\mathcal{F}| \geq T_{\geq 1}(n-1, k-1, t) \geq T_{\geq 1}(n-1, k-1, t-1) + 1 \geq n - k + t,$$

where the second last inequality follows from the fact that $T_{\geq r}(n, k, t)$ is an increasing function of t and the last inequality follows from the induction hypothesis.

Finally, when $k < t \leq 2^k - 1$, we are to show that $T_{\geq 1}(n, k, t) \geq T_{\geq 2}(n, k, t - k) + n$. We again use induction on k and assume that $\mathcal{F}$ is a minimal $\text{Turán-}_{\geq 1}(n, k, t)$ -system satisfying (6). If $[n] \subset \mathcal{F}$, then there is nothing to prove. We may therefore assume that there exists $a \in [n]$ such that $\{a\} \notin \mathcal{F}$. As before, $\mathcal{F}$ is a $\text{Turán-}_{\geq 1}(n-1, k-1, t)$ -system on $[n] \setminus \{a\}$. Note that this is only possible for $t \leq 2^{k-1} - 1$ since if $t \geq 2^{k-1}$, then $|\mathcal{F}(S)| \leq 2^{k-1} - 1 < t$ for any k -set S containing a . We may therefore apply the induction hypothesis to conclude that

$$|\mathcal{F}| \geq T_{\geq 1}(n-1, k-1, t) \geq T_{\geq 2}(n-1, k-1, t - k + 1) + (n-1).$$

Using the facts that $T_{\geq r}(n, k, t)$ is an increasing function of t and that $T_{\geq r}(n-1, k-1, t) \geq T_{\geq r}(n, k, t)$ (a $\text{Turán-}_{\geq r}(n-1, k-1, t)$ -system over $[n-1]$ is trivially a $\text{Turán-}_{\geq r}(n, k, t)$ -system over $[n]$), we finally have

$$|\mathcal{F}| \geq T_{\geq 2}(n-1, k-1, t - k) + 1 + (n-1) \geq T_{\geq 2}(n, k, t - k) + n. \quad \square$$

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